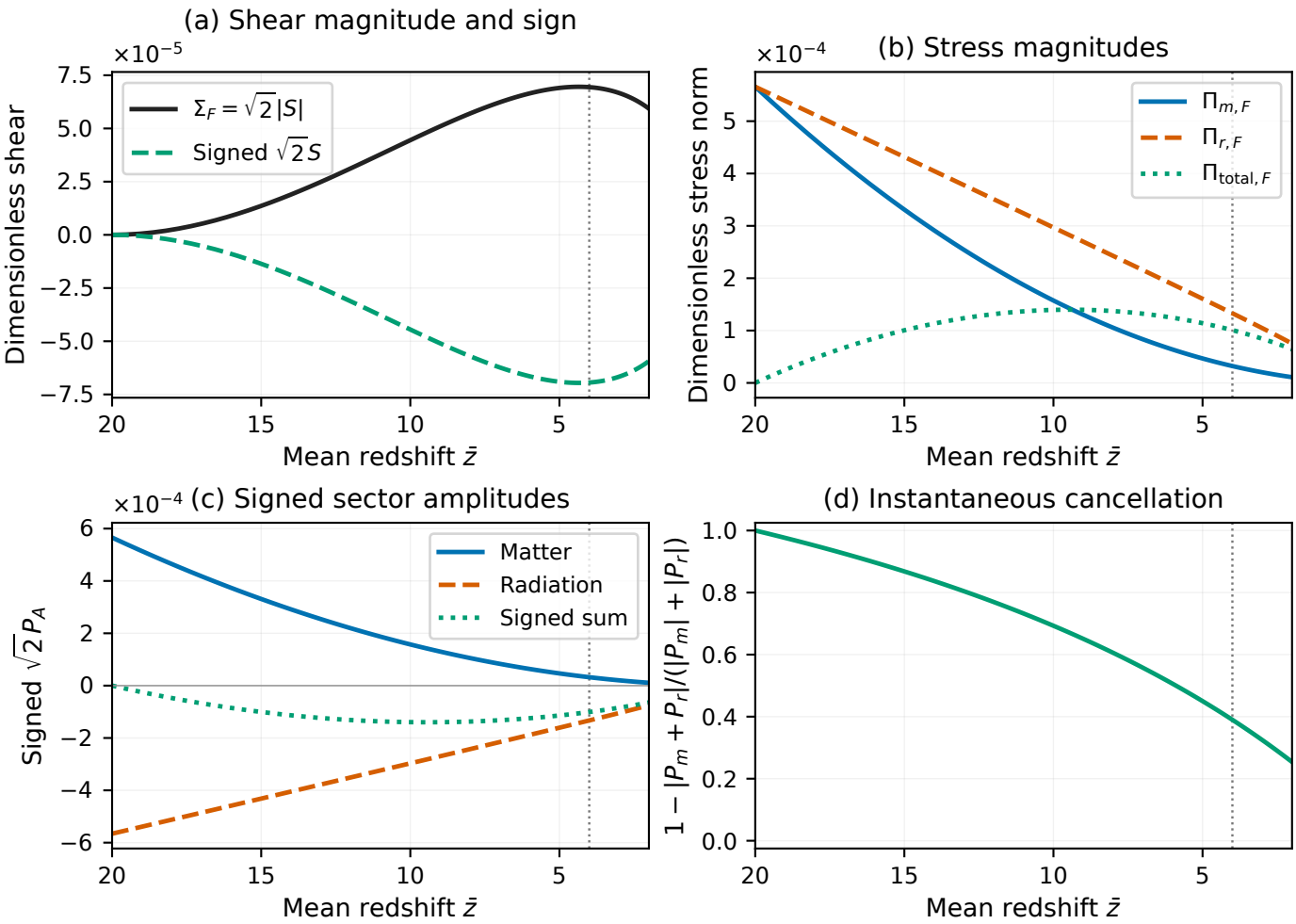
Separate matter and radiation stresses
(a) Shear magnitude and sign
×10⁻⁵
Dimensionless shear
Σ_F = √2|S|
Signed √2 S
Mean redshift z̄
(b) Stress magnitudes
×10⁻⁴
Dimensionless stress norm
Π_m,F
Π_r,F
Π_total,F
Mean redshift z̄
×10⁻⁴ (c) Signed sector amplitudes
Signed √2 P_A
Matter
Radiation
Signed sum
Mean redshift z̄
(d) Instantaneous cancellation
1 − |P_m + P_r|/(|P_m| + |P_r|)
Mean redshift z̄
Collisionless finite-owner LRS Bianchi I · same normal frame · z̄ = 20 → 2
Full Frobenius = √2 × half-contraction; case mixed_s+0_m+1; dotted line: continuation at z=4